

Supplementary Material

Sensor-Family Composition of the In-Situ Archive Changes the Apparent Bias and RMSD of Satellite Soil-Moisture Products

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S1 Archive composition and per-reference residuals

Table S1 gives the full manufacturer-family composition of the retained 2020–2023 ISMN archive and the mean ISMN–reference residual for each family against all three reference products, with 95 % bootstrap confidence intervals. The HydraProbe-family ERA5-Land CI does not overlap the Decagon/METER TDR CI, and the family swing ($0.17 \text{ m}^3 \text{ m}^{-3}$) is an order of magnitude larger than the within-family disagreement among references.

Table S1: ISMN 2020–2023 composition and mean ISMN–reference residual ($\text{m}^3 \text{ m}^{-3}$) by manufacturer family, 95 % bootstrap CI (5,000 resamples). Residual means use the valid-coverage matchup set; the composition count and the ERA5 residual count differ because not every station-sensor has valid coverage in every product (e.g., 1,231 of 1,274 HydraProbe-family sensors have valid ERA5-Land coverage).

Sensor family	n	Share (%)	ERA5-Land	SMAP L3.E	SMAP-SMI
HydraProbe family	1,274	78.4	$-0.097 [-0.103, -0.091]$	$-0.025 [-0.035, -0.016]$	$-0.037 [-0.043, -0.032]$
Decagon/METER TDR	95	5.8	$+0.068 [+0.028, +0.109]$	$+0.006 [-0.041, +0.057]$	$-0.074 [-0.103, -0.042]$
Campbell CS6xx	65	4.0	$-0.065 [-0.088, -0.042]$	$+0.019 [-0.001, +0.039]$	$-0.024 [-0.050, +0.005]$
ThetaProbe	59	3.6	$-0.039 [-0.061, -0.017]$	$+0.001 [-0.025, +0.027]$	$-0.001 [-0.028, +0.025]$
TRIME	2	0.1	—	—	—
other	130	8.0	$-0.056 [-0.079, -0.031]$	$+0.003 [-0.026, +0.033]$	$-0.022 [-0.046, +0.002]$
Total	1,625	100.0			

S2 Per-archive dielectric-free CRNS attribution

The pooled cross-check reported in the main text (92 CRNS $\leftrightarrow$ ISMN pairs, Welch’s $t = +4.07$, $p = 2.3 \times 10^{-4}$) decomposes by archive as follows. *COSMOS-USA* (reprocessed CRNS–ERA5-Land bias table of Zheng et al. [1], matched within 0.2° , $n = 56$): HydraProbe $+0.142 \text{ m}^3 \text{ m}^{-3} [+0.114, +0.170]$ vs non-HydraProbe $-0.067 [-0.174, +0.037]$; Welch’s $t = +3.61$, $p = 1.4 \times 10^{-3}$. *COSMOS-Europe* [2] (matched within 0.1° , $n = 31$): HydraProbe $+0.058 [+0.038, +0.078]$ vs non-HydraProbe $+0.010 [-0.016, +0.040]$; $t = +2.60$, $p = 0.019$. *COSMOS-UK* [3] on-site CRNS–TDT (same-footprint dielectric probe, $n = 49$): $+0.035 [+0.010, +0.061]$, excluded from the pooled within-archive family contrast because its design has no non-HydraProbe channel. *CosmOz Australia* [4] ($n = 5$): CRNS–HydraProbe $+0.162$ ($n = 2$), CRNS–Campbell CS6xx -0.043 ($n = 3$)—directionally consistent but too small for an independent test. Leave-one-archive deletion: dropping USA gives

$t = +3.91$, $p = 6.3 \times 10^{-4}$; dropping Europe $t = +4.05$, $p = 3.7 \times 10^{-4}$; dropping Australia $t = +3.68$, $p = 7.8 \times 10^{-4}$. Figure S1 summarizes the archive-level contrasts.

The COSMOS-USA row is a synthesized residual: the Zheng et al. CRNS–ERA5-Land bias is differenced against the station-matched ISMN–ERA5-Land residual from this work. The two ERA5-Land overlays are not redistributed as common-cell grids in this compact review package, so the USA row should be interpreted as a supporting rather than sole-identifying contrast. The USA-excluded leave-one-archive result above is therefore the conservative check used in the main text.

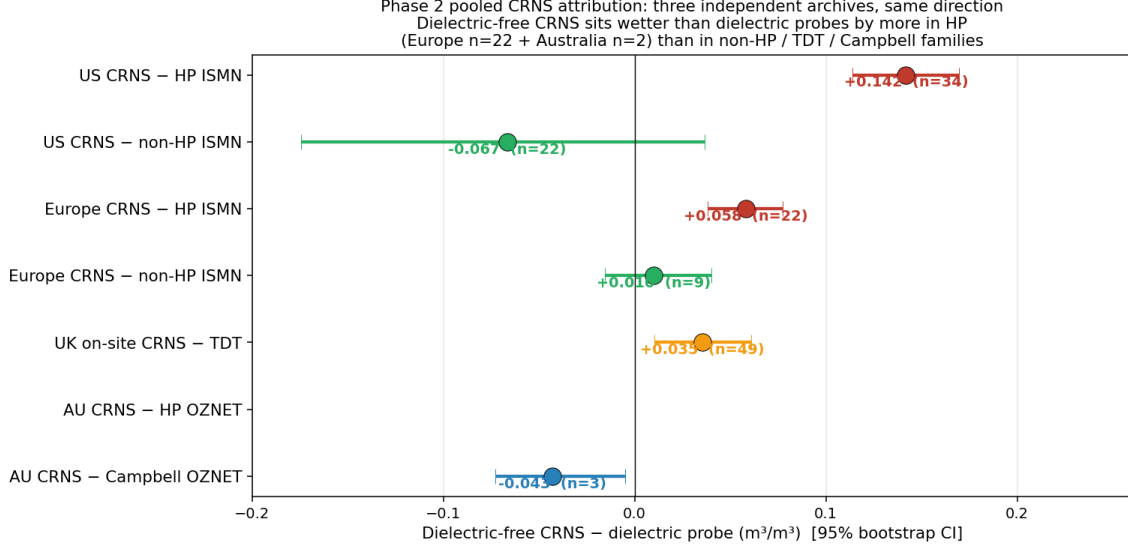


Figure S1: Per-archive dielectric-free CRNS attribution. Each row is the mean (CRNS – dielectric probe) residual for one archive-family comparison; error bars are 95 % bootstrap CI. The neutron reference is systematically wetter than the HydraProbe family and closer to non-HydraProbe families across three continents; Australia is shown for direction only.

S3 Near-surface constraint for the paired-sensor test

The main-text paired test uses $n = 21$ stations that host both a HydraProbe-family and a non-HydraProbe sensor in the same ISMN grid cell; both members are in the retained near-surface layer (depth-from ≤ 0.15 m). Table S2 reports the paired family contrast used in the main text. The inference is therefore restricted to co-located near-surface pairs.

Table S2: Paired-sensor offset (non-HydraProbe – HydraProbe) for co-located near-surface pairs.

Pair subset	pairs	mean diff (m³m ⁻³)	t	p
All co-located pairs	21	+0.075 [+0.031, +0.119]	+3.19	4.7×10^{-3}

S4 Real 2021 SMAP temporal validation check

The main letter focuses on mean-state bias and RMSD. To connect the family-composition result to standard satellite calibration/validation language, we also computed a real overpass-matched daily

matchup for 2021 using raw ISMN `.stm` hourly files and SMAP L3 SM_P descending/AM granules. The filters were ISMN quality flag "G", depth-from ≤ 0.15 m, recommended-quality SMAP retrievals, and at least 20 matched days per station-sensor. The resulting table contains 785 station-sensors and 78,600 matched station-days.

Table S3: Real 2021 overpass-matched temporal metrics by manufacturer family against SMAP L3 SM_P (AM). Bias is ISMN minus SMAP; values are station-sensor medians; brackets give 95 % network-cluster bootstrap CI. ThetaProbe ($n = 3$) is shown for completeness only.

Family	Stations	Bias	ubRMSE	R	Anomaly R
HydraProbe	637	-0.032 [-0.099, -0.017]	0.062 [0.051, 0.074]	0.317 [0.150, 0.443]	0.329 [0.088, 0.465]
Campbell CS6xx	58	-0.107 [-0.151, +0.099]	0.058 [0.032, 0.064]	0.166 [0.087, 0.622]	0.140 [-0.036, 0.657]
Decagon/METER TDR	57	+0.058 [+0.031, +0.139]	0.048 [0.028, 0.068]	0.622 [0.441, 0.647]	0.423 [0.264, 0.635]
other	30	-0.071 [-0.085, -0.033]	0.078 [0.056, 0.080]	0.055 [0.001, 0.436]	0.059 [-0.016, 0.283]
ThetaProbe	3	+0.195	0.030	0.639	0.608

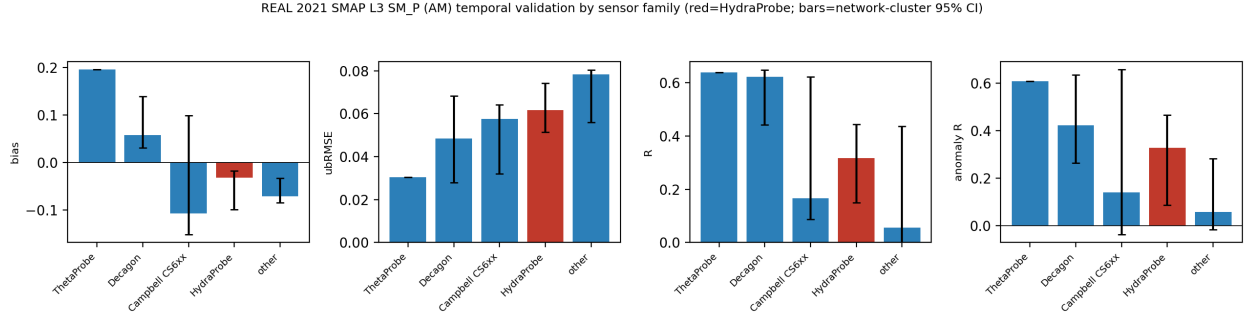


Figure S2: Real 2021 overpass-matched SMAP L3 SM_P (AM) temporal validation by sensor family. Bars are family medians and error bars are network-cluster 95 % CI.

The HydraProbe-versus-non-HydraProbe contrast is -0.032 versus $-0.036 \text{ m}^3 \text{ m}^{-3}$ for bias ($p = 9.8 \times 10^{-3}$), 0.062 versus $0.057 \text{ m}^3 \text{ m}^{-3}$ for ubRMSE ($p = 8.2 \times 10^{-4}$), 0.317 versus 0.273 for R ($p = 0.70$), and 0.329 versus 0.179 for anomaly R ($p = 0.055$). Because this check uses the 36-km SMAP SM_P product, one year, AM overpasses only, and point-to-grid matchups, it is interpreted as a standard-metric diagnostic rather than as a replacement for the mean-state SPL3SMP_E result in the main text. The pooled non-HydraProbe bias masks family heterogeneity: Decagon/METER TDR is wetter than HydraProbe ($+0.058$ versus $-0.032 \text{ m}^3 \text{ m}^{-3}$), whereas Campbell CS6xx and other records are drier (-0.107 and $-0.071 \text{ m}^3 \text{ m}^{-3}$).

S5 Temporal-window invariance

Because SMAP L3_E was extracted for 2023 only while ERA5-Land and SMAP-SMI span 2020–2023, we recomputed the ERA5-Land swvl1 (0–7 cm) grand mean for (i) the full 2020–2023 window and (ii) 2023 only, using an identical 0.1° nearest-neighbour protocol. The HydraProbe-family mean residual changes by $0.004 \text{ m}^3 \text{ m}^{-3}$ (3.9 %) between the two windows; all five families change by $\leq 0.012 \text{ m}^3 \text{ m}^{-3}$; and the driest→wettest family ranking (HydraProbe → Decagon/METER → other → ThetaProbe → Campbell CS6xx) is identical in both windows. The reported family ordering is therefore not an artifact of the ERA5 2020–2023 versus SMAP L3_E 2023-only window choice.

S6 Soil-texture control

Soil organic carbon (SOC; g kg^{-1}) was log-transformed before regression. A single ordinary-least-squares regression of the pooled three-reference mean residual on $\log_{10}(\text{SOC})$ gives a slope of $-0.078 \text{ m}^3 \text{ m}^{-3}$ per decade ($p = 1.7 \times 10^{-8}$, $n = 786$). At face value this reads as an organic-matter effect, but it is not robust to texture control. In a linear mixed-effects model with a random intercept for ISMN network and fixed-effect covariates for SoilGrids 2.0 [5] clay fraction and bulk density, sensor depth-from, and $|\text{latitude}|$, implemented with Statsmodels [6], the $\log_{10}(\text{SOC})$ coefficient collapses in all three references (ERA5-Land -0.029 , $p = 0.11$; SMAP L3_E -0.009 , $p = 0.77$; SMAP-SMI $+0.030$, $p = 0.08$), while the clay coefficient is highly significant in every reference ($p < 10^{-9}$). A leave-one-network jackknife of the OLS slope stays negative across all networks (-0.074 ± 0.011 ; range -0.096 to -0.034). The apparent organic-matter dependence is thus confounded with clay fraction and with HydraProbe deployment geography, consistent with a sensor-family/texture confounder rather than a direct organic-matter effect; the sensor-family ordering itself is unchanged by these controls.

References

- [1] Y. Zheng *et al.*, "Evaluation of reanalysis soil moisture products using cosmic ray neutron sensor observations across the globe," *Hydrol. Earth Syst. Sci.*, vol. 28, pp. 1999–2022, 2024, doi:10.5194/hess-28-1999-2024.
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- [6] S. Seabold and J. Perktold, "Statsmodels: econometric and statistical modeling with Python," in *Proc. 9th Python in Science Conf.*, 2010.